Sales Analysis, Insights and Forecasting Report

Overview

This analysis is based on the provided sales dataset and has been created using Power BI for visualization and Google Colab for forecasting. The objective is to understand overall sales performance, product contribution, sales trends over time, and future sales behavior.

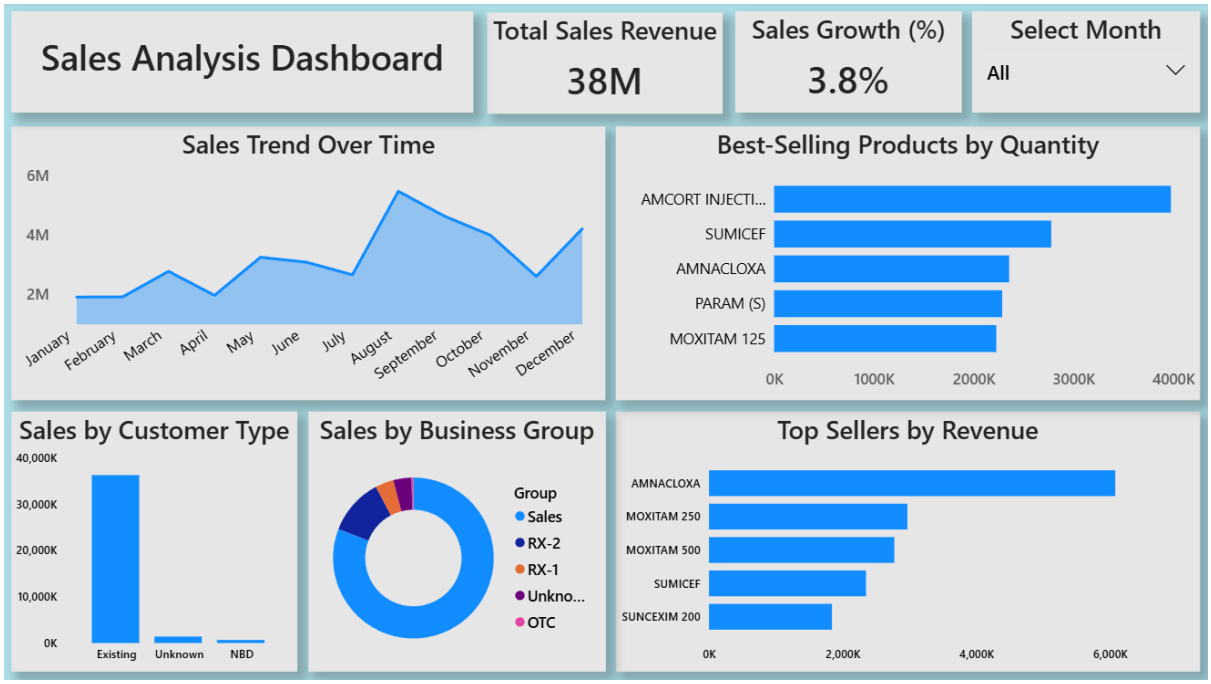

Figure 1: Power BI dashboard showing overall sales performance and trends.

Overall Sales Performance

- **Total Sales Revenue:** 38 Million
- **Overall Sales Trend:** Mixed, with ups and downs across different months
- **Sales Growth Percentage:** 3.8%

Observation:

Although sales fluctuate on a monthly basis, the overall business shows positive growth, indicating stable performance.

Sales Trend Over Time

Sales do not follow a straight upward or downward pattern. Some months perform better while others show a decline. This mixed trend suggests possible seasonality or changes in demand across different periods.

Product Performance

- **Best-selling product by quantity:** AMCORT INJECTION
- **Top-selling product by revenue:** AMNACLOXA

Observation:

Products with the highest sales volume are not always the highest revenue contributors. Both high-volume and high-value products play an important role in total sales.

Period-wise Analysis Using Month Slicer

Apr - Jun 2021

- Total Revenue: ₹2 Million
- Sales Growth: 93.4%
- Best-selling by quantity: AMCORT INJECTION
- Top-selling by revenue: AMNACLOXA

Observation:

This period shows very high growth, indicating a strong increase in demand during these months.

Jan - Dec 2022

- Total Revenue: ₹14 Million
- Sales Growth: 0.0%
- Best-selling by quantity: AMCORT INJECTION
- Top-selling by revenue: AMNACLOXA

Observation:

Sales during 2022 remained stable. Compared to earlier periods, growth slowed down, indicating that the business entered a steady phase.

Sales Forecasting (Future Trend)

Sales forecasting was performed using historical monthly sales data in Google Colab. A simple trend-based approach was used to project future sales.

The forecast shows a steady and linear growth trend, suggesting that if current patterns continue, sales are expected to remain stable in the coming months. This forecast can help in planning inventory, budgeting, and setting future sales targets. Actual results may vary based on market conditions.

Key Business Insights

- The business generated a total revenue of ₹38 million.
- Sales trends vary over time, showing both growth and decline in different months.
- High-volume products drive quantity, while premium products drive revenue.
- Stable performance in later periods highlights the importance of efficiency and consistency.
- Forecasting provides useful direction for future planning and decision-making.

Conclusion

Overall, the business shows stable performance with some fluctuations over time. By focusing on top-performing products and understanding low-performing periods, the company can improve future sales planning and business strategy.